

Comprehensive Performance Evaluation of Bowtie2, Minimap2, and GMAP for Short and Medium Read Alignments

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Abstract

Aligning DNA sequences to a reference genome is a fundamental task in bioinformatics, crucial for understanding genetic variations, identifying mutations, and exploring genomic structures. This study evaluates the performance of three widely used DNA sequence aligners: Bowtie2, Minimap2, and GMAP. Using artificial DNA samples of 300 bp and 500 bp aligned to the human genome GRCh38, we assess each tool based on alignment accuracy, speed, and resource utilization. Bowtie2 demonstrated balanced performance, offering efficient memory usage and moderate alignment accuracy, making it suitable for resource-limited environments. Minimap2 emerged as the most versatile aligner, excelling in alignment accuracy and computational efficiency across varying read lengths, positioning it as the optimal choice for high-throughput sequencing applications. GMAP, while achieving high alignment accuracy, showed significant memory usage and lacked multithreading support during indexing, thus limiting its applicability to specialized tasks where accuracy is prioritized over speed and resource efficiency. The evaluation was conducted using a standardized workflow with Bash and Python scripts, ensuring reproducibility and efficiency. This comprehensive benchmarking study underscores the importance of understanding the strengths and limitations of each aligner to optimize the accuracy and efficiency of genomic analyses.

Keywords: Bowtie2, Minimap2, GMAP, DNA Alignment, Alignment Time, CPU Usage, Memory Usage, Alignment Rate, Accuracy, Error Rate, Human Genome GRCh38, Artificial DNA Samples

1 Introduction

1.1 Importance of DNA Sequence Alignment

DNA sequence alignment is a fundamental task in bioinformatics, integral to various genomic analyses. By aligning DNA sequences to a reference genome, researchers can identify genetic variations, mutations, and structural differences. This process is crucial for understanding the genetic basis of diseases, exploring genomic structures, and performing functional genomics analyses. The accurate alignment of

sequences underpins many downstream applications, including variant detection, gene expression analysis, and genome assembly.

1.2 Challenges in Alignment

The alignment of DNA sequences presents several challenges. First, the volume of data generated by modern sequencing technologies is immense, requiring aligners to manage large datasets efficiently. Aligners must balance processing time and memory usage to handle these data volumes effectively. Second, the diversity in read length and error profiles adds complexity to the alignment process. Different sequencing technologies produce reads with varying characteristics, such as short reads with low error rates and long reads with higher error rates. Consequently, aligners must be adaptable to these variations to achieve high-quality results. Third, computational efficiency is a significant concern. Aligners need to perform tasks within practical timeframes while minimizing resource consumption. Finally, alignment accuracy is paramount for reliable downstream analyses. Tools must handle mismatches, insertions, deletions, and, in the case of RNA sequencing, complex splicing events accurately.

1.3 Study Objectives

This study evaluates three widely used DNA sequence aligners: Bowtie2, Minimap2, and GMAP. Bowtie2 is known for its speed and efficiency, especially with short reads. It builds on the success of its predecessor, Bowtie, by introducing support for gapped alignments and paired-end reads. Minimap2, developed for long-read sequencing technologies, excels in handling long reads and high error rates. It employs advanced algorithms and heuristics to ensure accuracy and speed. GMAP, specifically designed for spliced alignments, is particularly effective for RNA sequencing applications, handling complex genomic structures like introns and exons.

The primary objective of this study is to provide a comprehensive evaluation of these three aligners by aligning artificial DNA samples of 300 base pairs (bp) and 500 bp to the human genome reference GRCh38. The performance of each aligner is assessed based on several key metrics: alignment rate, alignment time, memory usage, accuracy, and error rate. By comparing these metrics, the study aims to

identify the strengths and limitations of each tool, offering insights into their suitability for various sequencing tasks.

To ensure the reliability and reproducibility of the results, a standardized workflow was employed. This workflow, implemented using Bash and Python scripts, automates the alignment process, data post-processing, and statistical analysis. The use of automation not only enhances efficiency but also ensures that the comparison between tools is conducted under consistent conditions. The study's methodology allows for a detailed and objective evaluation of each aligner's performance, providing potentially valuable information for researchers seeking to select the most appropriate tool for their needs.

1.4 Structure of the Report

The report is organized into several sections to facilitate a clear and comprehensive presentation of the study's findings. Following this introduction, the background and related work section reviews existing DNA alignment tools, with a focus on Bowtie2, Minimap2, and GMAP. The methodology section details the experimental setup, including data preparation and alignment procedures. The evaluation section presents the results and analysis of the alignment metrics, while the conclusion summarizes the findings and offers recommendations based on the comparative analysis.

2 Background and Related Work

DNA sequence alignment is a critical component of genomic research, facilitating the comparison of sequencing reads against a reference genome. The goal of alignment is to determine the best match between a query sequence and the reference genome, accounting for possible mismatches, insertions, and deletions. Various algorithms and tools have been developed to address this task, each with unique features and optimizations suited for different types of sequencing data and research objectives.

2.1 Bowtie

Bowtie was one of the pioneering tools designed for the ultrafast alignment of short DNA sequences to large genomes. It utilizes the Burrows-Wheeler transform (BWT) and an FM-index, enabling it to align reads rapidly and efficiently with low memory usage. Bowtie is particularly known for its speed and its ability to handle large datasets, though it is limited to ungapped alignments.

Langmead et al. (2009) introduced Bowtie in response to the growing demand for rapid and memory-efficient alignments driven by the advent of next-generation sequencing technologies.[1] The tool's performance set a new standard, demonstrating that short-read alignments could be performed efficiently without extensive computational resources. Bowtie was compared against other aligners such as MAQ and SOAP, which also aimed to provide efficient

alignment but employed different optimization strategies and trade-offs.

2.2 Bowtie2

Bowtie2 is an advanced version of the original Bowtie aligner, designed to handle the challenges associated with aligning short reads. It employs a divide-and-conquer approach to alignment, utilizing an efficient search algorithm based on the BWT and the FM-index. The BWT compresses the reference genome into a space-efficient format, which enables fast and memory-efficient alignment. The FM-index, an index structure derived from the BWT, allows Bowtie2 to perform exact and approximate alignments quickly.

Bowtie2 extends the capabilities of its predecessor by supporting gapped alignments and paired-end reads. It uses a two-step alignment process: first, it performs an initial alignment with a fast but approximate search, followed by a more accurate refinement step. This approach balances speed and accuracy, making Bowtie2 suitable for a range of applications, from small-scale research projects to large-scale sequencing studies.

Bowtie2 was benchmarked against several contemporary aligners, including Bowtie, BWA, and SOAP2, demonstrating superior speed and memory efficiency while maintaining high alignment accuracy.[2] These comparisons highlighted Bowtie2's ability to align reads with higher sensitivity and fewer mismatches, making it a preferred choice for many sequencing projects.

2.3 Minimap2

Minimap2, developed for fast pairwise alignment of nucleotide sequences, excels in aligning long reads, which are often produced by third-generation sequencing technologies such as PacBio and Oxford Nanopore. Unlike tools optimized for short-read alignment, Minimap2 is designed to handle the high error rates and substantial length variations characteristic of long reads. It employs a two-piece affine gap cost model and various heuristics to improve accuracy and speed. Notably, the Z-drop heuristic prevents the alignment of unrelated sequences between adjacent anchors, enhancing alignment quality. Additionally, Minimap2 uses SIMD (Single Instruction, Multiple Data) vectorization to accelerate computation, making it a highly efficient tool for long-read alignment.

Li (2018) highlighted Minimap2's versatility, showing its effectiveness in tasks ranging from read overlapping and genome assembly to spliced alignment for RNA sequencing data.[3] Its ability to perform well across different datasets and alignment tasks has made it a popular choice in the bioinformatics community.

2.4 GMAP

GMAP (Genomic Mapping and Alignment Program) is specifically tailored for spliced alignments, making it highly suitable for RNA sequencing data where reads span exon-intron boundaries.

GMAP utilizes a sophisticated approach to handle spliced alignments by incorporating a scoring system that distinguishes between true splicing events and sequencing errors. It aligns reads across splice junctions by considering the complex structure of eukaryotic genomes, including introns and exons. The aligner employs a dynamic programming algorithm that integrates a detailed model of genomic splicing. This model allows GMAP to accurately map reads to the reference genome, taking into account the presence of introns and exons. While GMAP is highly effective for spliced alignments, it is also more computationally intensive compared to other aligners, requiring significant memory and processing power.

Wu and Watanabe (2005) demonstrated that GMAP could accurately map cDNA sequences to the genome, identifying splice sites with high precision.[4] This capability is crucial for transcriptomics studies where understanding the splicing patterns of genes is essential.

2.5 Comparative Benchmarking

While Bowtie2 was compared with aligners like Bowtie, BWA, and SOAP2 in its foundational studies, our project aims to explore the limits of Bowtie2 by benchmarking it against tools not previously used in such comparisons: Minimap2 and GMAP. These tools represent different approaches to sequence alignment, tailored to the specific needs of various sequencing technologies and applications. Therefore, the performance of these alignment tools can vary significantly based on the specific requirements of a sequencing project.

This benchmarking analysis focuses on their performance in aligning artificial samples of different lengths (300 bp and 500 bp) to the human genome GRCh38, providing insights into their strengths and limitations. The aim of this comparative study is to help determine the most suitable aligner for different sequencing applications, especially as sequencing technologies continue to evolve and produce increasingly complex datasets.

3 Method

For this study, two artificial DNA samples of lengths 300 bp and 500 bp were generated. These samples were designed to mimic typical reads that might be encountered in various sequencing projects. The artificial reads were then aligned to the human genome reference sequence GRCh38 using Bowtie2, Minimap2, and GMAP. The alignment process and subsequent statistical analysis were automated using a Bash script for the alignment tasks and a Python script for the analysis and visualization of the results.

3.1 Reference Genome

The study utilized the GRCh38.p13 human genome assembly as the reference genome. GRCh38.p13, the thirteenth release of the Genome Reference Consortium's human genome, represents the most recent and comprehensive version, incorporating various updates to enhance accuracy and completeness. This reference genome was downloaded from the NCBI FTP server and served as the basis for subsequent indexing and alignment processes by the three alignment tools.

3.2 Synthetic Data Generation

To evaluate the performance of Bowtie2, Minimap2 and GMAP, synthetic paired-end sequencing data were generated. The synthetic reads were created using the wgsim tool, which simulates sequencing reads with specified characteristics. Two sets of paired-end reads were generated with different lengths: 500 base pairs (bp) and 300 bp. For each read length, a total of 10,000 read pairs were simulated. The mutation rates for substitutions and indels were set to 0.001, reflecting typical mutation rates observed in human populations, and an error rate of 0.01 was introduced. The quality of these synthetic reads was assessed using FastQC, which provided an overview of various quality metrics to ensure the validity of the simulated data.

3.3 Alignment Procedure and Modes

Bowtie2, Minimap2 and GMAP were employed for aligning the synthetic reads to the GRCh38.p13 reference genome. The alignment process involved two primary steps: indexing and read alignment. The reference genome was indexed by each aligner to create a searchable index, which was subsequently used for read alignment.

Bowtie2 supports both global and local alignment modes. In global alignment, which is the default mode, the tool aligns the entire length of the sequences, whereas local alignment (-local) focuses on aligning the most similar regions of the sequences. For each read length (500 bp and 300 bp), both global and local alignments were performed.

Minimap2 supports sensitive mode and assembly mode. Sensitive mode (-ax sr) is designed for high-throughput sequencing data, providing high accuracy for typical read lengths, while assembly mode (-ax asm5) accommodates longer sequences with up to 5% divergence. Minimap2 was employed to align the 300 bp reads using sensitive alignment mode only, whereas for the 500 bp reads both sensitive and assembly mode alignments were conducted.

GMAP alignment was employed using for both read lengths the -f sampe flag, which is appropriate for paired-end reads.

3.4 Data Processing

For each alignment tool, the alignment results in SAM format were processed using SAMtools. The processing steps included converting SAM files to BAM format, sorting the

BAM files, indexing the sorted BAM files, and generating alignment statistics. These steps ensure that the alignment data are in a format suitable for downstream analysis and allow for a comprehensive evaluation of the alignment performance across different tools.

3.5 Statistical Analysis and Visualization

The statistical analysis and visualization of the alignment results were performed using a Python script. This script parsed the alignment log files and the samtools statistics output to extract key performance metrics for each aligner. The log files from Bowtie2, Minimap2, and GMAP provided details on user time, system time, elapsed time, CPU usage, and maximum resident set size. The samtools statistics output for each aligner included information on the number of raw total sequences, mapped reads, unmapped reads, properly paired reads, mismatches, error rate and average quality. From these metrics, the alignment rate and accuracy were then calculated for each sample.

The Python script generated bar charts to illustrate the performance of each aligner. All metrics were plotted to provide a comprehensive comparison between Bowtie2, Minimap2, and GMAP. The script utilized libraries such as pandas for data manipulation, matplotlib and seaborn for visualization, and numpy for numerical operations.

3.6 Experimental Setup

The experiments were conducted on a high-performance computing cluster equipped with extensive resources to support large-scale alignment tasks. The cluster configuration included 7 CPU cores, 60 GB of RAM, and a 4 TB storage volume. This setup ensured that computational demands for the alignment tasks and subsequent analyses could be met effectively.

For each alignment tool, the alignment of both the 300 bp and 500 bp synthetic reads was performed separately. The alignment processes utilized multi-threading (configured to 7 threads) to optimize performance, and the resulting statistics were aggregated for analysis.

4 Evaluation

4.1 Performance Metrics for Bowtie2, Minimap2 and GMAP

For Bowtie2, Minimap2 and GMAP, the performance metrics were captured during the index-building and alignment processes (and stored in three separate log files, which were then processed by the Python script).

Bowtie2's performance metrics show significant variations between its indexing and alignment processes, as well as between its global and local alignment modes. During the index-building phase (step 0 in Tables 1 and 2), Bowtie2 took approximately 81 minutes, with a user time of approximately 7.84 hours and a system time of 123.40 seconds. This phase

had a high CPU usage of 581% and a maximum resident set size of approximately 5.64 GB. The alignment phase showed different resource demands based on the alignment mode. The local alignment for the 500 bp reads (step 2 in Tables 1 and 2) took 68 seconds, showing a CPU usage of 664% and a maximum resident set size of 4.45 GB, whereas for the 300 bp reads its execution (step 4 in Tables 1 and 2) was significantly faster (22 seconds). The global alignment, by contrast, required 19 seconds for the 500 bp reads (step 1 in Tables 1 and 2), with a CPU usage of 492% and a maximum resident set size of 3.85 GB, and only 7 seconds for the 300 bp reads (step 3 in Tables 1 and 2). This indicates that global alignments are less resource-intensive compared to local alignments. See Figures 1, 2 and 3 for a complete illustration.

Minimap2 demonstrated efficient performance across both indexing and alignment stages, across different alignment modes (sensitive and assembly). The index-building phase (step 0 in Tables 3 and 4) was completed in 143 seconds (approximately 2.4 minutes) with a user time of 158.29 seconds and a system time of 41.52 seconds. This phase exhibited a relatively low CPU usage of 139% and a maximum resident set size of approximately 12.02 GB. As expected, speed varied between the two modes. The sensitive alignment mode for the 500 bp reads (step 3 in Tables 3 and 4) took 24 seconds, with a CPU usage of 200% and a maximum resident set size of 7.93 GB, whereas for the 300 bp reads (step 2 in Tables 3 and 4) it required 20 seconds, with significantly less CPU usage (111%). The assembly mode (step 1 in Tables 3 and 4) took 20 seconds, with a CPU usage of 152% and a maximum resident set size of 8.05 GB. See figures 4, 5 and 6 for an illustration of Minimap2's efficient balance between speed and resource consumption.

GMAP exhibited the highest resource consumption among the tools evaluated. The index-building phase (step 0 in Tables 5 and 6) took approximately 137 minutes, with a user time of approximately 129.47 minutes and a system time of approximately 3.18 minutes, reflecting a CPU usage of 96% and a maximum resident set size of approximately 29.81 GB. Unlike Bowtie2 and Minimap2, GMAP does not support multithreading in the indexing step, which explains the extended time required. The alignment processes were also lengthy, taking approximately 5 minutes for the 300 bp reads (step 1 in Tables 5 and 6), with a CPU usage of 322% and a maximum resident set size of 6.98 GB. For the 500 bp reads, which took approximately 6.25 minutes, the CPU usage increased at 401% and the maximum resident set size at 7.64 GB. See figures 7, 8 and 9 for an illustration of GMAP's high computational and memory demands.

Overall, Minimap2 stands out for its efficient performance across both indexing and alignment stages, with relatively lower elapsed times and moderate memory usage compared to GMAP. Bowtie2, while competitive, showed considerable variability in performance, particularly between local and

Table 1. Time usage for Bowtie2's indexing and alignment commands

Step	Command	User Time (s)	System Time (s)	Elapsed Time (s)
0	bowtie2-build -threads 8 GCA_000001405.28_GRC...	28234.41	123.40	4876.0
1	bowtie2 -p 8 -x reference_index -1 output_read...	91.91	6.18	19.0
2	bowtie2 -p 8 -local -x reference_index -1 out...	447.28	5.27	68.0
3	bowtie2 -p 8 -x reference_index -1 output_read...	24.54	3.47	7.0
4	bowtie2 -p 8 -local -x reference_index -1 out...	133.26	3.52	22.0

Table 2. CPU and memory usage for Bowtie2's indexing and alignment commands

Step	Command	CPU Usage (%)	Max Resident Set Size (kB)
0	bowtie2-build -threads 8 GCA_000001405.28_GRC...	581.0	5912976
1	bowtie2 -p 8 -x reference_index -1 output_read...	492.0	4035548
2	bowtie2 -p 8 -local -x reference_index -1 out...	664.0	4666936
3	bowtie2 -p 8 -x reference_index -1 output_read...	395.0	3726768
4	bowtie2 -p 8 -local -x reference_index -1 out...	600.0	4001712

Table 3. Time usage for Minimap2's indexing and alignment commands

Step	Command	User Time (s)	System Time (s)	Elapsed Time (s)
0	minimap2 -t 8 -d reference_index_minimap2.mmi ...	158.29	41.52	143.0
1	minimap2 -ax asm5 -t 8 reference_index_minimap...	21.15	9.87	20.0
2	minimap2 -ax sr -t 8 reference_index_minimap2...	12.71	10.45	20.0
3	minimap2 -ax sr -t 8 reference_index_minimap2...	39.19	10.54	24.0

Table 4. CPU and memory usage for Minimap2's indexing and alignment commands

Step	Command	CPU Usage (%)	Max Resident Set Size (kB)
0	minimap2 -t 8 -d reference_index_minimap2.mmi ...	139.0	12606516
1	minimap2 -ax asm5 -t 8 reference_index_minimap...	152.0	8442892
2	minimap2 -ax sr -t 8 reference_index_minimap2...	111.0	8175460
3	minimap2 -ax sr -t 8 reference_index_minimap2...	200.0	8319452

global alignment modes, and high resource usage at the indexing step. The global alignment mode of Bowtie2 was significantly faster and less resource-intensive compared to the local alignment mode. GMAP's high computational and memory demands, coupled with its lack of multithreading support for the indexing step, make it less suitable for resource-constrained environments. Minimap2's balanced resource usage and speed make it the most efficient aligner among the three.

4.2 Alignment Metrics for 300 bp and 500 bp Reads

For the 300 bp reads (Table 7), Minimap2 achieved the highest alignment accuracy at 98.93% (that is, an error rate of 1.07% with 64003 mismatches), demonstrating its robustness and precision. Bowtie2 showed an accuracy of 98.71%

in global alignment mode (error rate of 1.29%, 77577 mismatches), while in local mode its accuracy was slightly lower at 97.72% (error rate of 2.28%, 130324 mismatches). GMAP, despite being the most computationally demanding aligner, had a relatively low accuracy of 98.25% (error rate of 1.75%, 149935 mismatches). See figures 10 and 12 for a comparison of the aligners' error rates and mismatches, respectively.

In terms of mapping efficiency, Minimap2 excelled by mapping all 20004 reads, with 20000 properly paired. Bowtie2 mapped 20003 reads in its local alignment mode, with 12652 properly paired. The alignment rate in its global mode was slightly lower (99.91%), mapping 19985 reads with 9986 properly paired. GMAP, while achieving an alignment rate of 100%, did not report any properly paired reads.

For the 500 bp reads (Table 8), Minimap2 again outperformed the other tools, with its sensitive mode achieving an

Table 5. Time usage for GMAP's indexing and alignment commands

Step	Command	User Time (s)	System Time (s)	Elapsed Time (s)
0	<code>gmap_build -d GRCh38 -D genome_index_gmap GCA_...</code>	7768.28	190.59	8230.0
1	<code>gmap -d GRCh38 -D genome_index_gmap -f sampe -...</code>	950.45	8.97	297.0
2	<code>gmap -d GRCh38 -D genome_index_gmap -f sampe -...</code>	1497.45	10.60	375.0

Table 6. CPU and memory usage for GMAP's indexing and alignment commands

Step	Command	CPU Usage (%)	Max Resident Set Size (kB)
0	<code>gmap_build -d GRCh38 -D genome_index_gmap GCA_...</code>	96.0	31256544
1	<code>gmap -d GRCh38 -D genome_index_gmap -f sampe -...</code>	322.0	7314484
2	<code>gmap -d GRCh38 -D genome_index_gmap -f sampe -...</code>	401.0	7908876

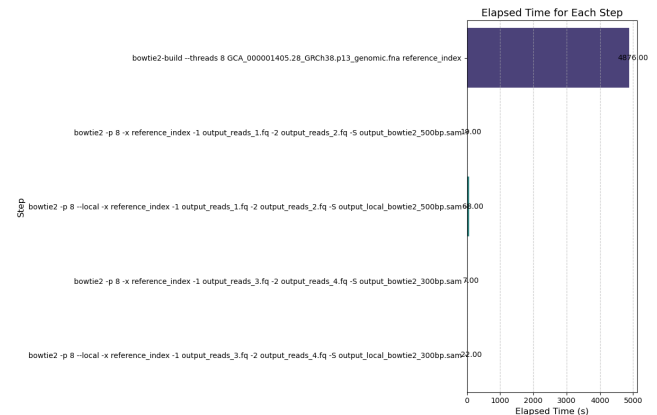
accuracy of 98.93% (error rate of 1.07%, 107066 mismatches) and its assembly mode reaching an accuracy of 99.02% (error rate of 0.98%, 97123 mismatches). The effectiveness of Minimap2's assembly mode for longer reads is illustrated in Figures 11 and 13, which compare the aligners' error rates and mismatches, respectively. Bowtie2 maintained high accuracy at 98.85% (error rate of 1.15%, 115022 mismatches) for the global alignment, though its local alignment was less effective with an accuracy of 96.95% (error rate of 3.05%, 270898 mismatches). GMAP's accuracy was again slightly low, at 98.13% (error rate of 1.87%, 264880 mismatches).

As far as mapping efficiency is concerned, Minimap2's sensitive mode mapped all 20004 reads, with 19908 properly paired, whereas its assembly mode mapped 19939 reads and did not report any properly paired reads. Bowtie2 also achieved an alignment rate of 100% in its local mode, with 14260 properly paired reads, although its alignment rate dropped at 99.92% in the global mode (mapping 19987 reads, with 10160 properly paired). GMAP's alignment rate was higher, although it failed again to report any properly paired reads.

When comparing the alignment metrics of Bowtie2, Minimap2 and GMAP between 300 bp and 500 bp reads, certain trends become apparent. Minimap2 consistently delivers the lowest error rates for both read lengths, indicating its robustness across varying read lengths. For 300 bp reads, Minimap2's accuracy was 98.93%, and it slightly improved to 99.02% for 500 bp reads in assembly mode, demonstrating its adaptability and precision for longer reads. Bowtie2 also performed well, though with some variability; its global alignment accuracy was 98.71% for 300 bp reads and 98.85% for 500 bp reads, showing it is well-suited for both short and long reads. However, its local alignment accuracy was lower for 500 bp reads (96.95%), indicating that local alignment mode may be less effective for longer reads. GMAP, while relatively accurate, showed a slight increase in error

rates from 0.018% for 300 bp reads to 0.019% for 500 bp reads, suggesting that it may be less efficient for longer reads.

Overall, Minimap2 is highly effective for both short (300 bp) and longer (500 bp) reads, demonstrating superior alignment accuracy and efficiency, making it a robust choice for diverse read lengths and computational environments. Bowtie2 also performs well across different read lengths, particularly in global alignment mode, making it a viable option for resource-constrained environments. GMAP, while accurate, is more suited for tasks where computational resources are abundant and high memory usage can be accommodated.

**Figure 1.** Elapsed time (s) for Bowtie2's commands.

5 Limitations of the Study

5.1 Sample Size and Sequencing Depth

The study utilized specific read lengths (300 bp and 500 bp) and a defined sample size. While these parameters provided valuable insights into the performance of each aligner, they may not fully represent the diversity of sequencing data encountered in real-world applications. The results might vary with different read lengths, sequencing depths, and

Table 7. Comparison of alignment metrics for the 300 bp reads

Metric	Local Bowtie2	Bowtie2	Minimap2	GMAP
raw total sequences	20004.000000	20004.000000	20004.000000	28606.00000
reads mapped	20003.000000	19985.000000	20004.000000	28606.00000
reads unmapped	1.000000	19.000000	0.000000	0.00000
reads properly paired	12652.000000	9986.000000	20000.000000	0.00000
mismatches	130324.000000	77577.000000	64003.000000	149935.00000
error rate	0.022767	0.012939	0.010666	0.01754
average quality	20.000000	20.000000	20.000000	20.00000
alignment rate	0.999950	0.999050	1.000000	1.00000
accuracy	0.977233	0.987061	0.989334	0.98246

Table 8. Comparison of alignment metrics for the 500 bp reads

Metric	Local Bowtie2	Bowtie2	Sensitive Minimap2	Assembly Minimap2	GMAP
raw total sequences	20004.000000	20004.00000	20004.000000	20004.000000	28500.000000
reads mapped	20004.000000	19987.00000	20004.000000	19939.000000	28499.000000
reads unmapped	0.000000	17.00000	0.000000	65.000000	1.000000
reads properly paired	14260.000000	10160.00000	19908.000000	0.000000	0.000000
mismatches	270898.000000	115022.00000	107066.000000	97123.000000	264880.000000
error rate	0.030482	0.01151	0.010705	0.009834	0.018663
average quality	20.000000	20.00000	20.000000	20.000000	20.000000
alignment rate	1.000000	0.99915	1.000000	0.996751	0.999965
accuracy	0.969518	0.98849	0.989295	0.990166	0.981337



Figure 2. CPU usage (%) for Bowtie2’s commands.

sample complexities. Future studies could benefit from a broader range of sequencing parameters to generalize the findings.

5.2 Tool Configuration

The configuration settings for each aligner were chosen based on standard practices and available resources. However, different parameter settings could influence the performance metrics. For instance, the number of threads used in

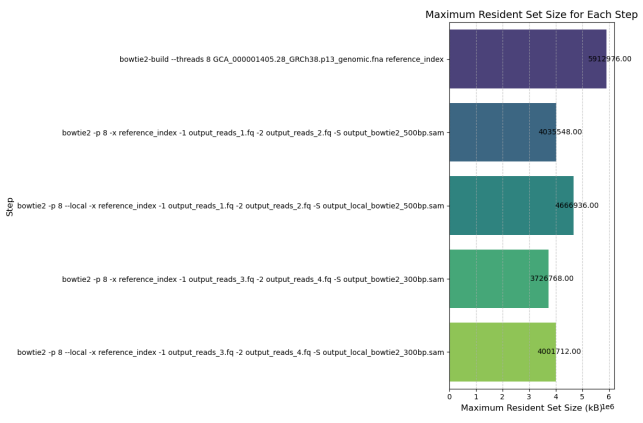


Figure 3. Maximum resident set size (kB) for Bowtie2’s commands.

Bowtie2 and Minimap2 was maximized based on the available CPU cores, which may not reflect typical usage scenarios. Additionally, GMAP’s lack of support for multithreading during indexing presents a limitation in direct comparison. Future evaluations could explore a wider range of configurations to better understand the impact of different settings on performance.

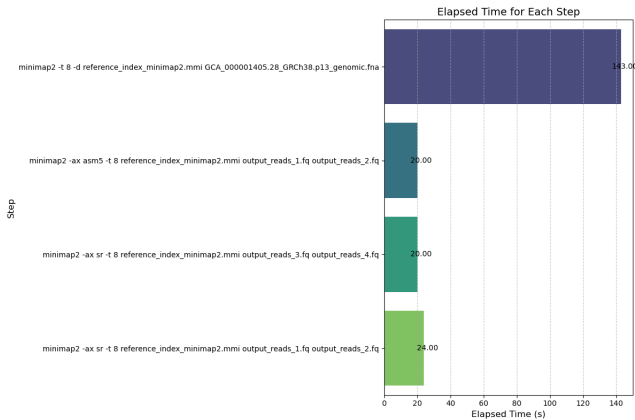


Figure 4. Elapsed time (s) for Minimap2’s commands.

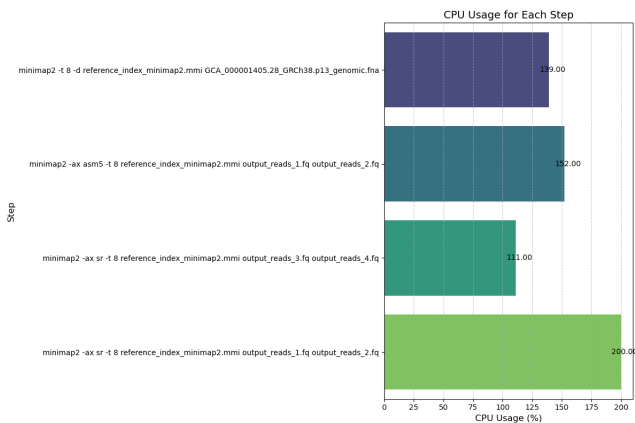


Figure 5. CPU usage (%) for Minimap2’s commands.

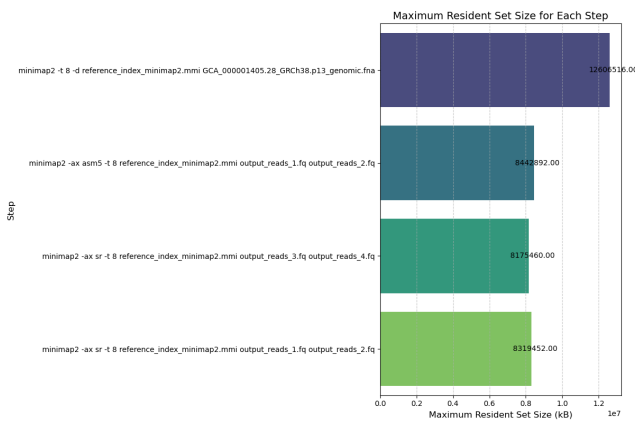


Figure 6. Maximum resident set size (kB) for Minimap2’s commands.

5.3 Resource Availability

The study was conducted on a high-performance computing cluster with substantial memory and CPU resources. While

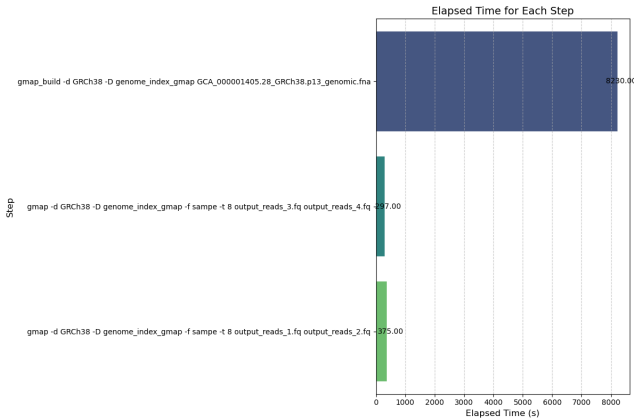


Figure 7. Elapsed time (s) for GMAP’s commands.

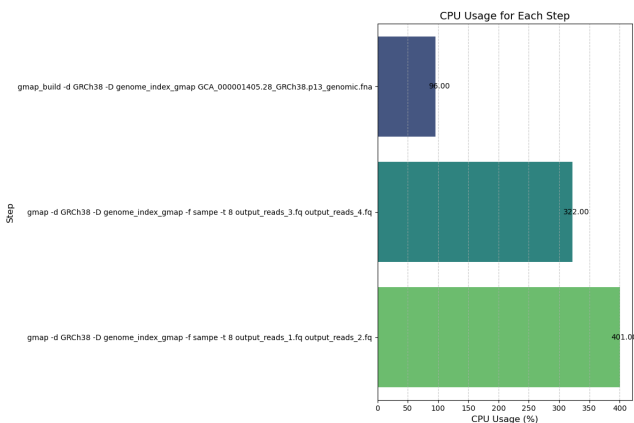


Figure 8. CPU usage (%) for GMAP’s commands.

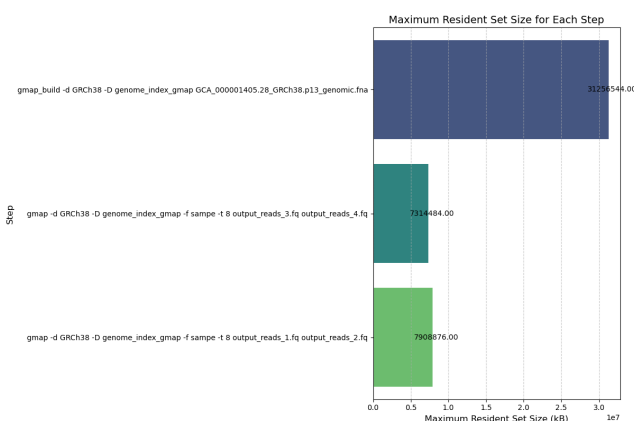


Figure 9. Maximum resident set size (kB) for GMAP’s commands.

this setup allowed for comprehensive testing, it may not accurately reflect the resource constraints in typical laboratory environments. The performance metrics, particularly for

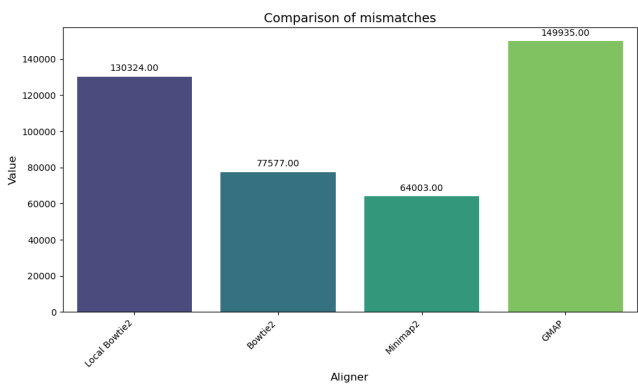


Figure 10. Comparison of alignment mismatches for the 300 bp reads.

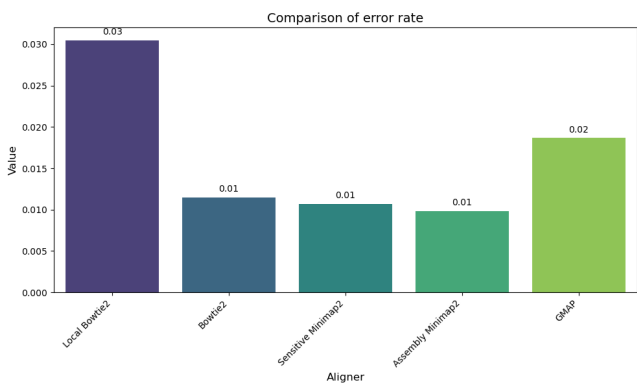


Figure 13. Comparison of alignment error rates for the 500 bp reads.

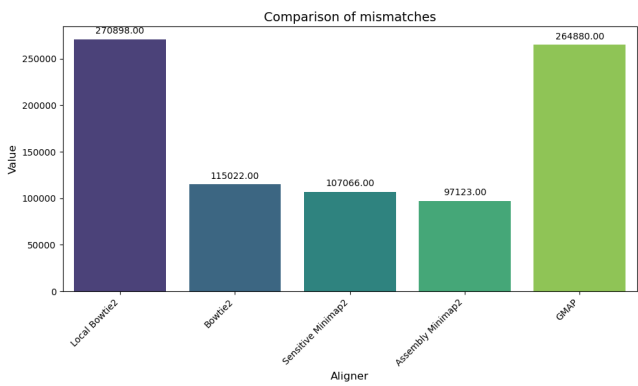


Figure 11. Comparison of alignment mismatches for the 500 bp reads.

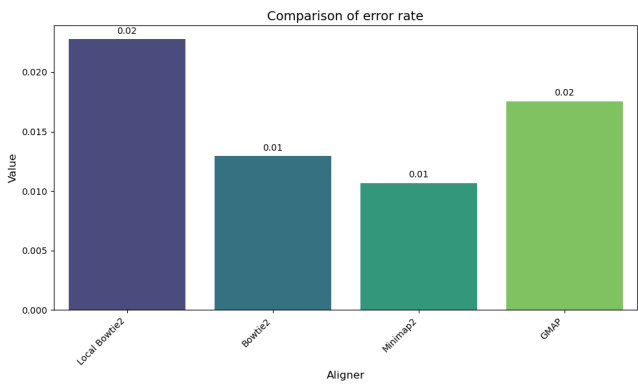


Figure 12. Comparison of alignment error rates for the 300 bp reads.

tools like GMAP, might differ significantly in less resource-rich settings. Future studies could include evaluations on a variety of hardware configurations to provide more generalizable recommendations.

6 Conclusion

This study conducted a comprehensive evaluation of three prominent DNA sequence aligners: Bowtie2, Minimap2, and GMAP. By aligning artificial DNA samples of 300 bp and 500 bp to the human genome GRCh38, we assessed each tool’s performance in terms of alignment accuracy, speed, and resource utilization. We implemented a standardized workflow using Bash and Python scripts to automate the alignment, post-processing, and statistical analysis, ensuring reproducibility and efficiency in handling large datasets and multiple alignment tasks. The Python script facilitated a detailed analysis of the alignment logs and statistics, enabling comprehensive performance comparisons and visualizations.

Bowtie2 demonstrated balanced performance with respect to speed and memory usage, excelling particularly in environments with limited computational resources. Although its alignment accuracy was slightly lower compared to Minimap2, Bowtie2’s efficiency in handling large datasets makes it a practical choice for many sequencing projects. Its performance indicates that it is well-suited for resource-limited environments where computational efficiency is paramount.

Minimap2 outperformed the other tools in alignment accuracy and overall efficiency, making it the most versatile aligner in this study. It showed robustness across varying read lengths and provided a good balance between computational resource usage and alignment performance. Minimap2’s comprehensive capabilities make it highly suitable for large-scale and diverse sequencing applications, positioning it as the optimal choice for high-throughput environments.

GMAP exhibited high alignment accuracy, particularly in complex reference genomes. However, its substantial memory requirements and lack of multithreading support during indexing limit its applicability in high-throughput environments. GMAP is best suited for specialized tasks where accuracy is prioritized over speed and resource efficiency. The performance of GMAP suggests it should be considered for

specialized, accuracy-critical tasks, provided that sufficient computational resources are available to handle its higher memory requirements.

Based on these findings, Bowtie2 is recommended for resource-limited environments due to its efficient use of computational resources and adequate performance across most metrics. Minimap2 is optimal for high-throughput and diverse applications, providing high accuracy and efficiency across different read lengths and sequencing tasks. GMAP should be considered for tasks that prioritize accuracy over speed and resource efficiency, particularly when dealing with complex reference genomes.

This study acknowledges several limitations, including the specific configurations used for each tool and the potential for varied performance with different sequencing parameters. Future research should explore a broader range of configurations, read lengths, and sequencing depths to generalize these findings further. Additionally, evaluations on varied hardware setups would provide more practical insights applicable to different laboratory environments.

In conclusion, the choice of an appropriate sequence aligner is crucial for the success of sequencing projects. By understanding the strengths and weaknesses of Bowtie2, Minimap2, and GMAP, researchers can make informed decisions

to optimize their alignment workflows. This study aims to contribute to the ongoing efforts to benchmark bioinformatics tools, ultimately aiding the scientific community in selecting the most effective tools for their specific needs.

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